

ImageLab Progress Report Week 4

CMPS4191: Advanced Web Technologies

University of Belize

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<https://github.com/andreshungbz/imagelab>

IMAGELAB	Title	ImageLab Progress Report Week 4
	Course	CMPS4191 Advanced Web Technologies
	Semester	2026-1
	Due Date	Sep 28, 2026

Progress Summary

The fourth and final week involved designing tests to measure job performance requirements. This also entailed verification of the functional requirements, the acceptance checklist, and the database constraints. Documentation was also added to reflect how the tests can be conducted.

Completed tasks included:

- Created a view in the database (v_metrics_jobs) for making querying for the job-record-related measurements more convenient. These include the Queue Wait, Processing Duration, and Job Duration measurements.
- Fixed a missing ON DELETE CASCADE for the jobs table consumer ID foreign key to make the down migrations work properly.
- Implemented a Python program to conduct the required measurement tests without the frontend. It handles the remaining Acknowledgment Latency, Polling Count, and Detection Delay measurements. Using a query to the created database view, the program aggregates all the required measurements into a printed table. The job count can be specified to measure 1 or 5 jobs.
- Wrote additional documentation in the README.md file detailing the tests and required dependencies.
- Miscellaneous updates to the developer tooling versions and Docker image versions.
- Preparation of the technical reflection document and demonstration.

The last requirements for the Week 4 Check-in were completed. None remained incomplete. In the lead-up to the final demonstration, smaller adjustments may be made as necessary.

One question that remains regards the optimal design of measurement tests. Upon analysis, the required measurements span both the database backend (Queue Wait, Processing Duration, and Job Duration) and the client side (Acknowledgment Latency, Polling Count, and Detection Delay). The decision was made to use scripting to take measurements since doing so with the direct frontend would be tedious and difficult to automate. A second decision was made to graduate from the shell script used in previous assessments to a Python program. While still possible with shell scripting, a full programming approach helps better organize and handle the required database queries, concurrent job requests, error handling, and print formatting. The design for the measurement tests clearly has multiple options.

The following page shows the measurement results for a single image and a five-image burst, with the API server running a 3-second image-processing delay. Of note are the increases in Queue Wait and Job Duration times, as well as in Polling Count, while the other measurements remain consistent.

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Measurement Tests

Single Image

```
λ just measure-baseline
uv run --directory ./python measurement --count 1 --api-url="http://localhost:$PORT" --db-dsn="$IMAGELAB_DB_DSN" --image-path="src/measurement/pittsburgh.jpg"
ImageLab Measurements (Images Count: 1)
=====
Job # | Public ID | Ack Latency | Queue Wait | Processing Duration | Job Duration | Polling Count | Detection Delay
-----|-----|-----|-----|-----|-----|-----|-----
1 | bb676206-7a50-40a4-9a83-16a092761874 | 24.7ms | 0.02s | 3.46s | 3.47s | 4 | 540.8ms
=====
```

Five-Image Burst

```
λ just measure-concurrent
uv run --directory ./python measurement --count 5 --api-url="http://localhost:$PORT" --db-dsn="$IMAGELAB_DB_DSN" --image-path="src/measurement/pittsburgh.jpg"
ImageLab Measurements (Images Count: 5)
=====
Job # | Public ID | Ack Latency | Queue Wait | Processing Duration | Job Duration | Polling Count | Detection Delay
-----|-----|-----|-----|-----|-----|-----|-----
1 | 51c7a25a-8419-4f4c-8461-d3edfda3b555 | 32.3ms | 14.09s | 3.46s | 17.54s | 18 | 524.8ms
2 | ecea731b-ae03-43da-862b-8ba74eebade | 31.3ms | 10.62s | 3.47s | 14.09s | 15 | 976.2ms
3 | 302a3e3f-93bf-4e79-b1a4-551dd538f147 | 27.8ms | 3.67s | 3.48s | 7.15s | 8 | 897.2ms
4 | c47f2cb1-1c3e-4469-9600-b97d50c1b1dd | 43.4ms | 0.19s | 3.48s | 3.67s | 4 | 340.6ms
5 | bc8a63b6-a144-4172-a43d-785b73422f33 | 28.1ms | 7.15s | 3.47s | 10.62s | 11 | 433.0ms
=====
```

Note that the ordering of the results may not be strictly increasing due to concurrent measurement execution.